

The first Hermite Gabor frame at rectangular density 4/7

A computer-assisted substantial partial result for arXiv:2403.10503

Autonomous mathematical discovery run `fa_banach_001`, lane 8

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Status. Candidate substantial partial result, likely valid. The proof is exact apart from a finite outward-rounded interval computation supplied in `code/verify_certificate.py`. It proves one new rational hyperbola of the still-open h_1 frame-set conjecture; it does not settle the full conjecture.

1 Source question and scope

Faulhuber, Shafkulovska, and Zlotnikov ask whether, for an odd Hermite function h_n , the Gabor system is always a frame when the lattice covolume is less than one and differs from $N/(N+1)$ [1, Introduction, p. 2]. They later state explicitly that h_1 is the only Hermite case for which the frame-set conjecture remains open [1, p. 19]. The two source excerpts are reproduced below.

which has not been treated in the literature so far ($h(\Lambda)$ is defined below in (1.2)). We will restrict our attention to the frame set of h_n for rectangular lattices of the form $\Lambda = a\mathbb{Z} \times b\mathbb{Z}$.

The case of the Gaussian function, i.e., $n = 0$, has been completely settled (for general point sets) independently by Lyubarskii [28] and in the articles of Seip [31] and Seip and Wallstén [32]. The case of the first Hermite function (and more generally odd functions) was treated by Lyubarskii and Nes [27] for rectangular lattices and extended to general lattices by one of the authors in [5].

In [15], Gröchenig came up with the following questions, which are now called the frame set conjecture for Hermite functions:

- For h_n with $n \in 2\mathbb{N}$, is the Gabor system $\mathcal{G}(h_n, \Lambda)$ always a frame when $\text{vol}(\Lambda) < 1$?
- For h_n with $n \in 2\mathbb{N} - 1$, is the Gabor system $\mathcal{G}(h_n, \Lambda)$ always a frame when $\text{vol}(\Lambda) < 1$ and $\text{vol}(\Lambda) \neq \frac{N}{N+1}$, $N \in \mathbb{N}$?

The questions were originally posed for rectangular lattices and are based on the restriction found in [27]. Due to the extension given in [5] they may be formulated for arbitrary lattices. However, the frame set conjecture for Hermite functions was answered negatively by Lemvig [25], who provided first counter-examples. More exceptions by Horst, Lemvig, and Videbaek [19] followed. A sufficiency criterion with numerical investigations was presented by Ghosh and Selvan [13].

This work is structured as follows. In Sec. 1.3 we introduce the general notation used in this work. In Sec. 2 we recall essential results from Gabor analysis in Sec. 2.1 provide

Figure 1: The frame-set questions in the source paper, PDF page 2.

For rectangular lattices $a\mathbb{Z} \times b\mathbb{Z}$, the conjecture specializes to

$$\mathcal{G}(h_1, a\mathbb{Z} \times b\mathbb{Z}) \text{ is a frame whenever } ab < 1, \quad ab \neq \frac{N}{N+1}.$$

$$\sum_{(k,l) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \mathcal{L}_n((n+1)\pi(k^2 + l^2)) e^{-\pi(n+1)(k^2 + l^2)/2} = -1.$$

The claims follow by adding the value at the origin, which contributes 1, to the sum. $\square$

For the proof of this result, we made use of the fact that the considered systems do not form frames. It would be nice to find a proof which does not need this fact.

As h_1 remains the only case where the frame set conjecture is still open (no further restrictions than the ones from [27] are known), we considered it worthwhile to apply the Janssen test in this case. The result is that we do not get any new insights.

Lemma 3.4. *For $\mathcal{G}(h_1, \delta^{-1/2}\mathbb{Z}^2)$, $\delta \geq 1$, the Janssen test quantity decreases, i.e.,*

$$\delta \mapsto \sum_{(k,l) \in \mathbb{Z}^2} |\mathcal{L}_1(\pi\delta(k^2 + l^2))| e^{-\pi\delta(k^2 + l^2)/2}$$

Figure 2: The source paper's explicit open-status statement for h_1 , PDF page 19.

The classical safety region is $ab < 1/2$. Lyubarskii and Nes proved the positive hyperbola $ab = 3/5$ and the negative hyperbolas $ab = N/(N+1)$ [2]. The theorem below adds the entire positive hyperbola $ab = 4/7$.

2 Proof intuition

At rational density p/q , the frame question is equivalent to full row rank of a finite Zak-transform matrix. For $p/q = 4/7$, two carefully chosen 4×4 minors cover the two halves of the fundamental x -interval. After centering every Zak series at its nearest Gaussian term, each minor has one determinant monomial whose Gaussian exponent is uniformly smaller than that of every competitor. A rigorous interval calculation at the square lattice shows that the total absolute mass of *all* monomials is less than 1.078 times the dominant one. Consequently the remainder has size less than 0.078 times the dominant monomial, so cancellation cannot make the minor vanish. Exact exponent gaps show that this ratio only decreases as the time spacing increases; Fourier symmetry supplies the complementary aspect ratios.

3 The result

We use the harmless normalization

$$h_1(t) = te^{-\pi t^2};$$

multiplying the window by a nonzero constant does not affect the frame property.

Theorem 1. *For every $a, b > 0$ with $ab = 4/7$, the rectangular Gabor system*

$$\mathcal{G}(h_1, a\mathbb{Z} \times b\mathbb{Z}) = \{e^{2\pi i b n(\cdot)} h_1(\cdot - am) : m, n \in \mathbb{Z}\}$$

is a frame for $L^2(\mathbb{R})$.

3.1 The rational rank criterion

For $L > 0$, write

$$Z_L g(u, \omega) = \sum_{k \in \mathbb{Z}} g(u - Lk) e^{2\pi i k L \omega}.$$

Lyubarskii and Nes prove that if $ab = p/q$ in lowest terms, then $\mathcal{G}(g, a\mathbb{Z} \times b\mathbb{Z})$ is a frame if and only if the $p \times q$ matrix

$$P_{s,t}(x, \omega) = Z_{aq} g\left(x + at + \frac{aq}{p}s, \omega\right), \quad 0 \leq s < p, \quad 0 \leq t < q, \quad (1)$$

has rank p throughout a fundamental rectangle [2, Theorem 2]. It is enough below to treat $0 \leq x < a/p$ and all ω , because the matrix is quasiperiodic.

3.2 Two minors and their dominant terms

Set $p = 4$, $q = 7$, $y = x/a$, and first assume

$$a \geq a_0 := \sqrt{4/7}. \quad (2)$$

Thus $0 \leq y \leq 1/4$. On $0 \leq y \leq 1/8$, select columns (1, 4, 5, 6) of (1); on $1/8 \leq y \leq 1/4$, select columns (0, 1, 4, 6). After shifting the arguments by integer multiples of $7a$, the centered dimensionless arguments $a(y + B_{sj})$ are given by

$$B_- = \begin{pmatrix} 1 & -3 & -2 & -1 \\ 11/4 & -5/4 & -1/4 & 3/4 \\ -5/2 & 1/2 & 3/2 & 5/2 \\ -3/4 & 9/4 & 13/4 & -11/4 \end{pmatrix} \quad (3)$$

on the first half, and

$$B_+ = \begin{pmatrix} 0 & 1 & -3 & -1 \\ 7/4 & 11/4 & -5/4 & 3/4 \\ -7/2 & -5/2 & 1/2 & 5/2 \\ -7/4 & -3/4 & 9/4 & -11/4 \end{pmatrix} \quad (4)$$

on the second. The determinant permutation is written as a tuple whose s th entry is the chosen column index in these displayed matrices. Choose

$$\sigma_- = (3, 2, 1, 0), \quad \sigma_+ = (0, 3, 2, 1).$$

The monomial obtained from these permutations and the centered term of each Zak series will be called the dominant monomial. Its magnitude is

$$D_{\pm}(a, y) = a^4 \prod_{s=0}^3 |y + (B_{\pm})_{s, \sigma_{\pm}(s)}| \exp\left(-\pi a^2 \sum_{s=0}^3 (y + (B_{\pm})_{s, \sigma_{\pm}(s)})^2\right). \quad (5)$$

Every factor in (5) has absolute value at least $1/8$ on its assigned half interval, so $D_{\pm}(a, y) > 0$.

For a selected entry with centered coefficient $c = y + B_{sj}$, put

$$A_{sj}(a, y) = a \sum_{k \in \mathbb{Z}} |c - 7k| e^{-\pi a^2 (c - 7k)^2}, \quad T_{\pm}(a, y) = \text{per}(A(a, y)).$$

Expanding both the determinant and every Zak series shows that $T_{\pm}$ is the sum of the absolute values of all determinant monomials, including the chosen monomial $D_{\pm}$. Hence, for every ω ,

$$|\det P_{\pm}(x, \omega)| \geq D_{\pm}(a, y) - (T_{\pm}(a, y) - D_{\pm}(a, y)) = 2D_{\pm}(a, y) - T_{\pm}(a, y). \quad (6)$$

Lemma 1 (Certified dominance). *On the respective half intervals,*

$$\frac{T_{\pm}(a_0, y)}{D_{\pm}(a_0, y)} < 1.077024 < 2.$$

Moreover, $T_{\pm}(a, y)/D_{\pm}(a, y)$ is nonincreasing for $a \geq a_0$.

Proof. The first assertion is a finite, outward-rounded interval certificate. Each half interval is divided into 512 equal rational boxes. On each box the script evaluates all 24 permanent terms using 50-decimal interval arithmetic. Five translates, $-2 \leq k \leq 2$, are evaluated directly. Since $|c| \leq 27/8 < 7/2$, the omitted two-sided tail is bounded by

$$2 \left(Ae^{-\kappa A^2} + \frac{e^{-\kappa A^2}}{14\kappa} \right), \quad \kappa = \frac{4\pi}{7}, \quad A = \frac{35}{2}, \quad (7)$$

which is less than $6.015 \cdot 10^{-238}$ per entry. The largest interval upper endpoint for T/D is

$$1.077023594634533$$

(rounded upward), occurring in the box adjacent to $y = 1/8$ on either side.

For the monotonicity assertion, remove the common a^4 from every ratio. Exact rational arithmetic verifies that, among centered terms, every non-dominant permutation has sum-of-squares exponent at least $7/2$ above the dominant permutation. Also, replacing any centered coefficient by a different 7-translate increases its square by at least

$$49 - 14 \max |c| = 49 - 14 \cdot \frac{27}{8} = \frac{7}{4}.$$

Thus every non-dominant monomial divided by the dominant monomial has the form $Ce^{-\pi a^2 \Delta}$ with $C \geq 0$ independent of a and $\Delta \geq 7/4$. Their sum $(T - D)/D$ is therefore nonincreasing in a . All rational gap checks, tail bounds, and interval evaluations are performed and asserted by the supplied verifier. $\square$

Proof of Theorem 1. Under (2), Lemma 1 and (6) show that the selected minor is nonzero on each half of $0 \leq y \leq 1/4$, uniformly in ω . Therefore the matrix (1) has rank four everywhere, and the Lyubarskii–Nes criterion proves the frame property.

If instead $a < a_0$, then $b = (4/7)/a > a_0$. The Fourier transform interchanges the time and frequency lattice parameters, and $\hat{h}_1 = -ih_1$ (up to the chosen normalization). Hence the system with parameters (a, b) is unitarily equivalent to the already treated system with parameters (b, a) . This proves the theorem for every aspect ratio. $\square$

4 Verification and novelty bounds

Run from the repository root:

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/partial/\
  2403.10503_h1_density_four_sevenths_frame/code/verify_certificate.py
```

With Python 3.11.14 and `mpmath` 1.3.0, the script prints the exact gaps $7/2$ and $7/4$, interval upper bounds 1.0770235946345321 on both halves, and PASS. The discrepancy between that printed float and the rounded display in the proof is in the safe direction. See `verification.md` for the full recorded output and review checklist.

The bounded novelty check searched the run registry and attempt indexes, the source paper’s citations, arXiv, and general web search using arXiv ids 2403.10503, 1108.2684, and 2605.26709, together with “first Hermite Gabor frame”, “rational density”, “ $4/7$ ”, and close variants. No theorem for $ab = 4/7$ was found. The May 2026 paper [3] explicitly says the region $1/2 < ab < 1$ for h_1 is still open and proves only that the Wirtinger criterion cannot enter it. This is a bounded search, not a guarantee of novelty.

5 Limitations and human-review recommendation

The full frame-set conjecture remains open. The present dominant-monomial method certifies $4/7$ and appears numerically effective at several other rational densities, but its absolute-value estimate fails at some high densities (already $13/15$), where phase cancellation among multiple leading monomials must be controlled instead.

A reviewer should check (i) the transcription of the rational rank criterion, (ii) the two centered matrices and selected permutations, (iii) that the reverse-triangle estimate counts every determinant/Zak monomial once, and (iv) the interval script’s outward rounding and tail formula. These are the only substantive dependencies.

References

- [1] M. Faulhuber, I. Shafkulovska, and I. Zlotnikov, *On the frame property of Hermite functions and exploration of their frame sets*, J. Fourier Anal. Appl. 31 (2025), article 21; [arXiv:2403.10503](#).
- [2] Y. Lyubarskii and P. G. Nes, *Gabor frames with rational density*, Appl. Comput. Harmon. Anal. 34 (2013), 488–494; [arXiv:1108.2684](#).
- [3] M. Faulhuber, *On a fundamental barrier of the Wirtinger criterion for Gabor systems with odd functions*, 2026; [arXiv:2605.26709](#).